Optimization

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Lesson 3

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Support Vector Machine

Objective

Support Vector Machine (SVM) is a recent and very effective class of supervised *Machine Learning* (ML) models and algorithms for **classification** and regression.

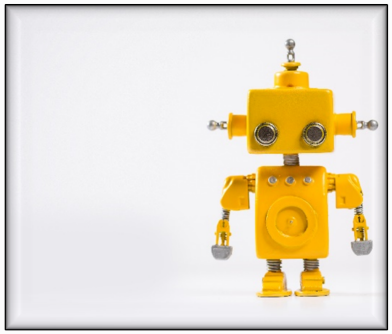
Origins

SVM were first introduced in the 70s:

- Vapnik, V., Chervonenkis, A., 1974. **Theory of Pattern Recognition**.

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Applications of Support Vector Machine



Robotics



Medical Diagnosis



Speech Recognition



Character Recognition



Spam Filtering

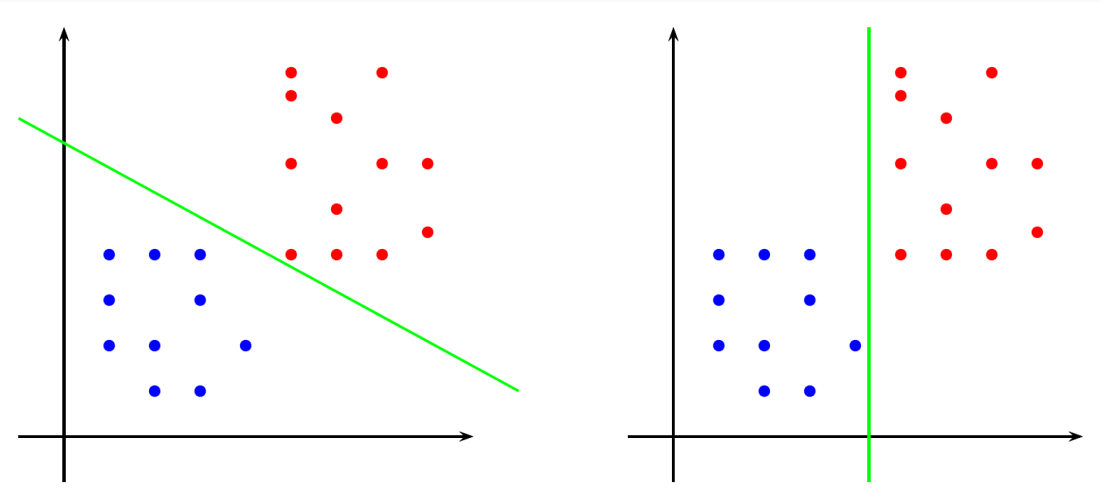


Nutrition

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Support Vector Machine

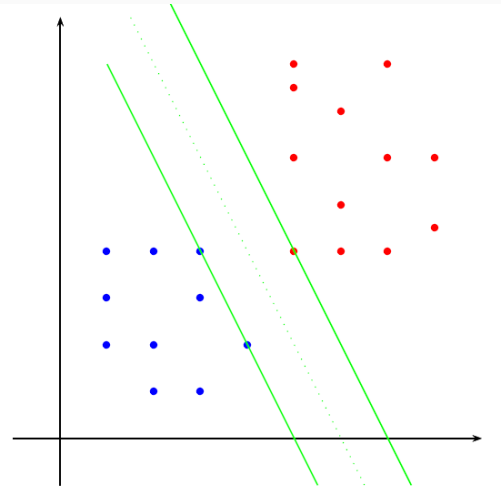
- The figure below shows two different hyperplanes (in this case, two lines) that both separate the red and blue points.
- In terms of training set the two hyperplanes are equivalent but the classification of new points is highly different.



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Support Vector Machine

- For classification problems, the fundamental idea in SVM is to determine, instead of a single separating hyperplane, **a pair of parallel hyperplanes for which the margin is maximum**.



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Linear Separable SVM Model

- Let $\{(x^p, y_p)\}$ with $p = 1, \ldots, P$ be the set of training data (*i.e.*, the **training set**) where:
  - $x^i \in \mathbb{R}^N$ represents an input observation,
  - and $y_p \in \{-1, +1\}$ its associated label.
- The training set is **linearly separable** if and only there exists an hyperplane $H_{w,\theta}$ where $w \in \mathbb{R}^N$ and $\theta \in \mathbb{R}$ such that:

$$\begin{cases} w^\top x^p + \theta > 0 & \text{if} \quad y_p = +1 \\ w^\top x^p + \theta < 0 & \text{if} \quad y_p = -1 \end{cases}$$

  for all $p = 1, \ldots, P$.

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Linear Separable SVM Model

- For a linearly separable training set there always exists a separating hyperplane $H_{\bar{w},\bar{\theta}}$ such that for all $p = 1, \ldots, P$:

$$\begin{cases} \bar{w}^\top x^p + \bar{\theta} \geq +1 & \text{if} \quad y_p = +1 \\ \bar{w}^\top x^p + \bar{\theta} \leq -1 & \text{if} \quad y_p = -1. \end{cases}$$

- In fact, let $H_{w,\theta}$ be a separating hyperplane and let:

$$\alpha = \min_{p=1,\ldots,P,\ y_p=+1} w^\top x^p + \theta$$

$$\beta = \max_{p=1,\ldots,P,\ y_p=-1} w^\top x^p + \theta.$$

  Note that $\alpha > 0$ and $\beta < 0$.

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Linear Separable SVM Model

- Moreover, let $\gamma = \min\{\alpha, -\beta\}$, $\bar{w} = \frac{w}{\gamma}$ and $\bar{\theta} = \frac{\theta}{\gamma}$.
- Now, for each $p = 1, \ldots, P$ if $y_p = +1$ then:

$$w^\top x^p + \theta \geq \gamma \quad \text{and hence} \quad \bar{w}^\top x^p + \bar{\theta} \geq 1$$

while if $y_p = -1$ then:

$$w^\top x^p + \theta \leq -\gamma \quad \text{and hence} \quad \bar{w}^\top x^p + \bar{\theta} \leq -1.$$

- Therefore, a training set is linearly separable if there exist a separating hyperplane $H_{w,\theta}$ such that for all $p = 1, \ldots, P$:

$$\begin{cases} w^\top x^p + \theta \geq +1 & \text{if} \quad y_p = +1 \\ w^\top x^p + \theta \leq -1 & \text{if} \quad y_p = -1. \end{cases}$$

- Note that conditions above can be rewritten as:

$$y_p(w^\top x^p + \theta) \geq 1 \quad \forall\, p = 1, \ldots, P.$$

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Linear Separable SVM Model

Definition (Hyperplane Margin)

Let the training set be linearly separable. The **margin** $\rho_{w,\theta}$ *of* $H_{w,\theta}$ *is the minimum distance between points in the training set and the hyperplane:*

$$\rho_{w,\theta} = \min_{p=1,\dots,P} \frac{|w^\top x^p + \theta|}{||w||}$$

where $||\cdot||$ *is the Euclidean norm.*

- SVM will choose as separating hyperplane H_{w^*,θ^*} the one for which the margin is maximum.
- That can be achieved by solving the following optimization problem:

$$\begin{array}{ll} \max & \rho_{w,\theta} \\ s.t. & y_p(w^\top x^p + \theta) \geq 1 \quad p = 1, \dots, P. \end{array}$$

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Linear Separable SVM Model

- Notice that since for all $p = 1, \dots, P$:

$$y_p(w^\top x^p + \theta) \geq 1 \quad \iff \quad |w^\top x^p + \theta| \geq 1$$

then from the definition of $\rho_{w,\theta}$ we have:

$$\rho_{w,\theta} = \min_{p=1,\dots,P} \frac{|w^\top x^p + \theta|}{||w||} \geq \frac{1}{||w||}.$$

- Moreover, for each separating hyperplane $H_{w,\theta}$ there exists a separating hyperplane $H_{\bar{w},\bar{\theta}}$ such that:

$$\rho_{w,\theta} \leq \rho_{\bar{w},\bar{\theta}} = \frac{1}{||w||}$$

and $p', p'' \in \{1, \dots, P\}$ such that

$$y^{p'} = +1 \quad \text{and} \quad \bar{w}^\top x^{p'} + \bar{\theta} = +1$$
$$y^{p''} = -1 \quad \text{and} \quad \bar{w}^\top x^{p''} + \bar{\theta} = -1.$$

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Linear Separable SVM Model

- In fact, for the separating hyperplane $H_{w,\theta}$ let p', $p'' \in \{1, \dots, P\}$ such that $y_{p'} = +1$, $y_{p''} = -1$ and:

$$\alpha = \min_{p=1,\dots,P,\ y_p=+1} \frac{|w^\top x^p + \theta|}{||w||} = \frac{|w^\top x^{p'} + \theta|}{||w||}$$

$$\beta = \min_{p=1,\dots,P,\ y_p=-1} \frac{|w^\top x^p + \theta|}{||w||} = \frac{|w^\top x^{p''} + \theta|}{||w||}.$$

- Note that:

$$\rho_{w,\theta} \leq \min\{\alpha, \beta\} \leq \frac{\alpha + \beta}{2} = w^\top \left(x^{p'} - x^{p''} \right).$$

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Linear Separable SVM Model (further calculations)

- Let now $\gamma, \delta \in \mathbb{R}$ such that:

$$\gamma w^\top x^{p'} + \delta = +1$$
$$\gamma w^\top x^{p''} + \delta = -1$$

that is:

$$\gamma = \frac{2}{w^\top \left(x^{p'} - x^{p''}\right)} \quad \text{and} \quad \delta = \frac{w^\top \left(x^{p'} - x^{p''}\right)}{w^\top \left(x^{p'} + x^{p''}\right)}.$$

- Now:

$w^\top x^{p'} + \theta \geq 1 \geq 0$ and $w^\top x^{p''} + \theta \leq -1$, hence $-\left(w^\top x^{p''} + \theta\right) \geq 1 \geq 0.$

- Therefore:

$$w^\top x^{p'} + \theta - \left(w^\top x^{p''} + \theta\right) = w^\top \left(x^{p'} - x^{p''}\right) \geq 2 > 0.$$

- So, $0 < \gamma \leq 2$.

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Linear Separable SVM Model (further calculations)

- Let $\bar{w} = \gamma w$ and $\bar{\theta} = \delta$. The Hyperplane $H_{\bar{w},\bar{\theta}}$ is a separating hyperplane. In fact, since:

$$w^\top x^p \geq w^\top x^{p'} \quad \forall\, p \in \{1, \dots, P : y_p = +1\}$$
$$w^\top x^p \leq w^\top x^{p''} \quad \forall\, p \in \{1, \dots, P : y_p = -1\}$$

  then

$$\gamma w^\top x^p + \bar{\theta} \geq \gamma w^\top x^{p'} + \bar{\theta} = +1 \quad \forall\, p \in \{1, \dots, P : y_p = +1\}$$
$$\gamma w^\top x^p + \bar{\theta} \leq \gamma w^\top x^{p''} + \bar{\theta} = -1 \quad \forall\, p \in \{1, \dots, P : y_p = -1\}.$$

- Hence:

$$\bar{w}^\top x^p + \bar{\theta} \geq +1 \quad \forall\, p \in \{1, \dots, P : y_p = +1\}$$
$$\bar{w}^\top x^p + \bar{\theta} \leq -1 \quad \forall\, p \in \{1, \dots, P : y_p = -1\}.$$

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Linear Separable SVM Model (further calculations)

- Moreover:

$$\begin{aligned}
\rho_{\bar{w},\bar{\theta}} &= \min_{p=1,\dots,P} \frac{|\bar{w}^\top x^p + \bar{\theta}|}{||\bar{w}||} \\
&= \frac{1}{||\bar{w}||} \\
&= \frac{1}{||\gamma w||} \\
&= w^\top \left(x^{p'} - x^{p''} \right) \\
&= \rho_{w,\theta}
\end{aligned}$$

and

$$\begin{aligned}
\bar{w}^\top x^{p'} + \bar{\theta} &= +1 \\
\bar{w}^\top x^{p''} + \bar{\theta} &= -1.
\end{aligned}$$

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Linear Separable SVM Model

Therefore the problem:

$$\begin{aligned} \max \quad & \rho_{w,\theta} \\ s.t. \quad & y_p(w^\top x^p + \theta) \geq 1 \quad p = 1, \dots, P \end{aligned}$$

is equivalent to solve the following convex quadratic optimization problem:

Hard Margin SVM

$$\begin{aligned} \min \quad & \frac{1}{2}||w||^2 \\ s.t. \quad & y_p(w^\top x^p + \theta) \geq 1 \quad p = 1, \dots, P. \end{aligned} \tag{*}$$

Notice that even though the objective function of problem (*) is not strictly convex (in w and θ), it admits a unique solution.

The **Lagrangian function** associated to problem (*) is:

$$\mathcal{L}(w, \theta, \alpha) = \frac{1}{2} w^\top w - \sum_{p=1}^{P} \alpha_p \left(y_p (w^\top x^p + \theta) - 1 \right)$$

and the Karush–Kuhn–Tucker optimality conditions are:

KKT Conditions

$$w = \sum_{p=1}^{P} \alpha_p y_p x^p$$

$$\sum_{p=1}^{P} \alpha_p y_p = 0$$

$$\alpha_p \geq 0 \qquad p = 1, \ldots, P$$

$$y_p (w^\top x^p + \theta) - 1 \geq 0 \qquad p = 1, \ldots, P$$

$$\alpha_p \left[y_p (w^\top x^p + \theta) - 1 \right] = 0 \qquad p = 1, \ldots, P.$$

Remark

- *The first KKT condition indicates that the optimal w is a linear combination (with coefficients $y_p\alpha_p$) of the training set elements x^p.*
- *The last set of complementarity conditions implies that the only vectors entering in the linear combination are those for which*

$$y_p(w^\top x^p + \theta) = 1.$$

These vectors are called support vectors.

Optimality Conditions and Dual Problem

The **Wolfe dual** is given by:

$$\begin{aligned} \max_{w,\theta,\alpha} \quad & \mathcal{L}(w,\theta,\alpha) \\ s.t. \quad & \nabla_w \mathcal{L}(w,\theta,\alpha) = 0 \\ & \nabla_\theta \mathcal{L}(w,\theta,\alpha) = 0 \\ & \alpha \geq 0. \end{aligned}$$

Now:

$$\begin{aligned} \mathcal{L}(w,\theta,\alpha) &= \tfrac{1}{2} w^\top w - \textstyle\sum_{p=1}^{P} \alpha_p \left(y_p (w^\top x^p + \theta) - 1 \right) \\ &= -\tfrac{1}{2} w^\top w + w^\top \left(w - \sum_{p=1}^{P} \alpha_p y_p x^p \right) - \theta \sum_{p=1}^{P} \alpha_p y_p + \sum_{p=1}^{P} \alpha_p. \end{aligned}$$

Therefore, after substituting for w, the dual problem can be written as:

$$\begin{aligned} \min_{\alpha} \quad & \frac{1}{2}\left(\sum_{p=1}^{P}\alpha_p y_p x^p\right)^\top \left(\sum_{q=1}^{P}\alpha_q y_q x^q\right) - \sum_{p=1}^{P}\alpha_p \\ s.t. \quad & \sum_{p=1}^{P}\alpha_p y_p = 0 \\ & \alpha_p \geq 0 \qquad\qquad p = 1, \ldots, P. \end{aligned}$$

Let now $K : \mathbb{R}^N \times \mathbb{R}^N \to \mathbb{R}$ with $K(\bar{x}, \hat{x}) = \bar{x}^\top \hat{x}$. Then, the minimization problem above can be re-written as:

$$\begin{aligned} \min_{\alpha} \quad & \tfrac{1}{2}\alpha^\top Q\alpha - e^\top \alpha \\ s.t. \quad & y^\top \alpha = 0 \\ & \alpha \geq 0 \end{aligned}$$

where e is a vector with all components equal to 1, $Q \in \mathbb{R}^{P\times P}$ and:

$$Q_{pq} = y_p y_q K(x^p, x^q).$$

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Optimality Conditions and Dual Problem

Once the dual problem is solved (let α^* be the optimal solution), the separating hyperplane

$$w^{*\top} x + \theta^* = 0$$

is obtained as follows:

- $w^* = \sum_{p=1}^{P} \alpha_p^* y_p x^p$ (see first KKT condition);
- if p' is a support vector (that is, $\alpha_{p'}^* > 0$), then:

$$y_{p'}(w^{*\top} x^p + \theta^*) - 1 = 0$$

(see last KKT conditions) from which θ^* can be easily computed.

Then, the decision function $h(\cdot)$ for linear SVM is:

Decision Function

$$h(x) = \text{sign}\left(w^{*\top} x + \theta^*\right) = \text{sign}\left(\sum_{p=1}^{P} \alpha_p^* y_p K(x^p, x) + \theta^*\right) \qquad (\diamond)$$

where $x \in \mathbb{R}^N$.

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EXERCISE

Consider the following training set in $\mathbb{R}^2$:

Point	Label
$x^1 = (1, 0)$	$y_1 = +1$
$x^2 = (0, 1)$	$y_2 = +1$
$x^3 = (-1, 0)$	$y_3 = -1$
$x^4 = (0, -1)$	$y_4 = -1$

We consider the hard-margin Support Vector Machine problem:

$$\begin{aligned} \min_{w,\theta} \quad & \frac{1}{2}\|w\|^2 \\ \text{s.t.} \quad & y_p(w^\top x^p + \theta) \geq 1, \quad p = 1, \dots, 4. \end{aligned}$$

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Answer the following questions:

1. Write the problem in standard inequality form
2. Construct the Lagrangian function.
3. Write the Karush–Kuhn–Tucker (KKT) conditions.
4. Derive the Wolfe dual problem.
5. Solve the dual problem.
6. Determine the optimal w and θ.
7. Identify the support vectors.
8. Compute the geometric margin of the classifier.

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